Prime Factors, HCF and LCM

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KEY WORDS factor *(tauwehe)* · prime number *(tau toitū)* · highest common factor *(tauwehe pātahi nui rawa)*

A Factors and primes ●○○

Remember: A prime number has exactly two factors: 1 and itself. The number 1 is not prime.

1. List all the factors of each number.

a) 12 _____ **b)** 20 _____

c) 17 _____ **d)** 18 _____

e) 30 _____ **f)** 36 _____

2. Write the first five multiples of each number.

a) 6 _____ **b)** 9 _____

c) 12 _____ **d)** 15 _____

3. Circle the prime numbers in each list.

a) 2 9 11 15 21 23 27 29

b) 1 31 33 37 39 41 49 51

B Products of prime factors ●●○

Watch out: Keep going until every factor is prime. $12 = 3 \times 4$ is not finished, because 4 is not prime: $12 = 2 \times 2 \times 3$.

4. Write each number as a product of its prime factors. A factor tree may help.

a) 6 **b)** 15 **c)** 12 **d)** 20

e) 18 **f)** 30 **g)** 42 **h)** 50

i) 28 **j)** 45 **k)** 66 **l)** 70

C Index form ●●○

Remember: When a prime factor repeats, use a power: $2 \times 2 \times 2 \times 3 = 2^3 \times 3$.

5. Write each number as a product of its prime factors, using powers.

a) 8 **b)** 24 **c)** 36 **d)** 40

e) 72 **f)** 100 **g)** 54 **h)** 200

6. Work out the number.

a) $2^2 \times 5$ **b)** 2×3^2 **c)** $2^3 \times 7$ **d)** $3^2 \times 5^2$

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D Highest common factor ●●○

> **Remember:** The HCF is the largest number that divides exactly into both numbers.

7. Find the highest common factor of each pair.

a) 6 and 9 **b)** 8 and 12 **c)** 10 and 15 **d)** 12 and 18

e) 14 and 21 **f)** 20 and 30 **g)** 16 and 24 **h)** 9 and 20

i) 24 and 36 **j)** 18 and 45 **k)** 28 and 42 **l)** 36 and 60

E Lowest common multiple ●●○

> ⚠ **Watch out:** Do not mix the two up. The HCF is never bigger than the smaller number. The LCM is never smaller than the bigger number.

8. Find the lowest common multiple of each pair.

a) 2 and 3 **b)** 4 and 6 **c)** 3 and 5 **d)** 6 and 9

e) 4 and 10 **f)** 5 and 6 **g)** 8 and 12 **h)** 3 and 12

i) 10 and 15 **j)** 9 and 12 **k)** 6 and 14 **l)** 12 and 20

F Problems ●●●

9. Sausages come in packs of 8 and buns come in packs of 6. What is the smallest number of packs of each that gives the same number of sausages and buns?

10. One bus leaves a stop every 6 minutes and another every 8 minutes. They leave together at 9:00 am. After how many minutes do they next leave together?

11. A teacher has 24 pencils and 36 rubbers. She makes identical packs that use all of them, with no items left over. What is the largest number of packs she can make?

★ Extension

12. List all the prime numbers between 20 and 30.

13. Find the HCF and the LCM of 12 and 18. Multiply them together. What do you notice?

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Prime Factors, HCF and LCM

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS factor *(tauwehe)* · prime number *(tau toitū)* · highest common factor *(tauwehe pātahi nui rawa)*

A Factors and primes ●○○

Remember: A prime number has exactly two factors: 1 and itself. The number 1 is not prime.

1. List all the factors of each number.

a) 10 _____ **b)** 28 _____

c) 13 _____ **d)** 24 _____

e) 40 _____ **f)** 16 _____

2. Write the first five multiples of each number.

a) 4 _____ **b)** 7 _____

c) 11 _____ **d)** 20 _____

3. Circle the prime numbers in each list.

a) 3 8 13 15 19 21 25 31

b) 1 27 37 43 45 47 57 59

B Products of prime factors ●●○

⚠ **Watch out:** Keep going until every factor is prime. $12 = 3 \times 4$ is not finished, because 4 is not prime: $12 = 2 \times 2 \times 3$.

4. Write each number as a product of its prime factors. A factor tree may help.

a) 10 **b)** 21 **c)** 18 **d)** 28

e) 12 **f)** 42 **g)** 30 **h)** 75

i) 44 **j)** 63 **k)** 78 **l)** 110

C Index form ●●○

Remember: When a prime factor repeats, use a power: $2 \times 2 \times 2 \times 3 = 2^3 \times 3$.

5. Write each number as a product of its prime factors, using powers.

a) 16 **b)** 20 **c)** 27 **d)** 60

e) 48 **f)** 90 **g)** 108 **h)** 400

6. Work out the number.

a) $2^2 \times 7$ **b)** 2×5^2 **c)** $2^3 \times 3$ **d)** $2^2 \times 3^2$

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D Highest common factor ●●○

> **Remember:** The HCF is the largest number that divides exactly into both numbers.

7. Find the highest common factor of each pair.

a) 4 and 6 **b)** 9 and 12 **c)** 10 and 25 **d)** 12 and 16

e) 15 and 21 **f)** 20 and 50 **g)** 18 and 24 **h)** 8 and 15

i) 24 and 40 **j)** 27 and 45 **k)** 30 and 42 **l)** 48 and 60

E Lowest common multiple ●●○

> ⚠ **Watch out:** Do not mix the two up. The HCF is never bigger than the smaller number. The LCM is never smaller than the bigger number.

8. Find the lowest common multiple of each pair.

a) 2 and 5 **b)** 4 and 6 **c)** 3 and 4 **d)** 6 and 8

e) 4 and 14 **f)** 5 and 7 **g)** 6 and 9 **h)** 4 and 12

i) 10 and 25 **j)** 8 and 12 **k)** 6 and 15 **l)** 12 and 18

F Problems ●●●

9. Sausages come in packs of 10 and buns come in packs of 4. What is the smallest number of packs of each that gives the same number of sausages and buns?

10. One bus leaves a stop every 9 minutes and another every 12 minutes. They leave together at 9:00 am. After how many minutes do they next leave together?

11. A teacher has 30 pencils and 45 rubbers. She makes identical packs that use all of them, with no items left over. What is the largest number of packs she can make?

★ Extension

12. List all the prime numbers between 30 and 40.

13. Find the HCF and the LCM of 20 and 30. Multiply them together. What do you notice?

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